

Percentage, Ratio & Profit/Loss — Level 2 — Answer Key

1. **C) 600** — A 25% increase multiplies by 1.25; a 20% decrease then multiplies by 0.80. Combined factor: $1.25 \times 0.80 = 1.00$ exactly — so the final value always equals the original. Original = 600.
2. **C) 168** — The largest and smallest parts differ by $7-3 = 4$ parts, equal to 48, so 1 part = 12. Sum of all parts = $(3+4+7) \times 12 = 14 \times 12 = 168$.
3. **B) 8:15** — Scale A:B = 2:3 to 8:12 ($\times 4$) and B:C = 4:5 to 12:15 ($\times 3$), so both share B = 12. Then A:B:C = 8:12:15, giving A:C = 8:15.
4. **C) 19%** — Marked price = $500 \times 1.40 = ₹700$. Selling price after 15% discount = $700 \times 0.85 = ₹595$. Profit = $595 - 500 = ₹95$. Profit % = $95/500 \times 100 = 19\%$.
5. **C) 10%** — The interest earned between year 3 and year 5 is $₹6,000 - ₹5,200 = ₹800$ over 2 years, so ₹400 per year. Principal = $₹5,200 - (3 \times ₹400) = ₹4,000$. Rate = $(400 \times 100)/4,000 = 10\%$.
6. **C) 300** — 20% of 30% of 5,000 = $0.20 \times 0.30 \times 5,000 = 300$.
7. **C) 30, 50** — Let the numbers be $3x$ and $5x$. $(3x+10)/(5x+10) = 2/3 \rightarrow 3(3x+10) = 2(5x+10) \rightarrow 9x+30 = 10x+20 \rightarrow x = 10$. Numbers = 30 and 50.
8. **C) ₹680** — Gain = 15% of the selling price = 15% of ₹800 = ₹120. Cost price = selling price - gain = $₹800 - ₹120 = ₹680$.
9. **C) 20 years** — Becoming 4 times itself means the interest earned equals 3 times the principal. Using $SI = (P \times R \times T) \div 100$ with interest = $3P$: $3P = (P \times 15 \times T) \div 100 \rightarrow T = 300 \div 15 = 20$ years.
10. **C) 300** — 25% more than 96 = $96 \times 1.25 = 120$. If 40% of the number is 120, the number = $120 \div 0.40 = 300$.
11. **C) ₹480** — If ₹456 represents a 5% loss, it is 95% of the cost price. Cost price = $456 \div 0.95 = ₹480$.
12. **C) 24, 30** — Let the ages be $4x$ and $5x$. $(4x+6)/(5x+6) = 5/6 \rightarrow 6(4x+6) = 5(5x+6) \rightarrow 24x+36 = 25x+30 \rightarrow x = 6$. Ages = 24 and 30.
13. **C) ₹1,800** — Time = 2 years 6 months = 2.5 years. $SI = (8,000 \times 9 \times 2.5) \div 100 = ₹1,800$.
14. **D) 56%** — Combined factor = $1.30 \times 1.20 = 1.56$, i.e. a 56% overall increase.
15. **C) ₹3,240** — Total cost = $₹2,400 + ₹300 = ₹2,700$. Profit = 20% of ₹2,700 = ₹540. Selling price = $2,700 + 540 = ₹3,240$.

How to use this: 10+/15 is strong for a Level-2-focused set. Q1 and Q14 are worth re-reading carefully even if you got them right — spotting that "+25% then -20%" cancels out exactly, while "+30% then +20%" does *not* simply add to 50%, is exactly the kind of percentage-chain trap Level 2 papers use.

